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# CS 621 DATA ANALYTICS

## Data Structures

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# LISTS AND DICTIONARIES

- Check materials covered in the Python Review Module.

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# TUPLES AND SETS

- Tuples are immutable lists.
  - t =('b','c','a','g','h')
- No duplicates in Sets.
  - x=set([1,2,3,1,3,5])
  - s1 = {2, 1, 3, 4, 4}, s2 = {5, 3, 2, 1, 1}
  - s1.union(s2)
  - s1.intersection(s2)
  - s1.difference(s2)

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# NUMPY ARRAYS

- import numpy as np
- l = [5.2, 1.3, 4.4, 6.7]
- a = np.array(l)
- numpy arrays are often used in applications involving heavy numerical computation.

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# PANDAS

- Pandas a data analysis and manipulation tool written in Python.
- Pandas is highly welcome among data scientists.
- https://pandas.pydata.org

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# TWO DATA STRUCTURES IN PANDAS

- A **Series** is one-dimensional array-like object containing an array of data and an associated array of data labels(index).

- A **DataFrame** represents a tabular, spreadsheet-like data structure.

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# SERIES

- Series is an ordered dictionary.
  - >>> from pandas import Series, DataFrame
- >>> obj2 = Series([4, 7, -5, 3], index=['d', 'b', 'a', 'c'])
- >>> obj2.index
- >>> obj2.values
- >>> obj2['a']
- >>> obj2[2]

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# SERIES CONTD

- >>> sdata = {'Ohio':35000, 'Texas':71000, 'Oregon': 16000, 'Utah':5000}
- >>> obj3 = Series(sdata)
- >>> obj3 .head(n) #show the first n objects

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# DATAFRAME CONTD

- A DataFrame represents a tabular, spreadsheet-like data structure.
- Often used with data loaded from csv files.

```
>>> from pandas import Series, DataFrame

>>> data = {'state':['Ohio', 'Ohio', 'Ohio', 'Nevada', 'Nevada'],
'year':[2000, 2001, 2002, 2001, 2002], 'pop':[1.5, 1.7, 3.6,
2.4, 2.9]}

>>> frame = DataFrame(data)

>>> frame
```

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# DATAFRAME CONTD

```
>>> DataFrame(data, columns=['year', 'state', 'pop'])

>>> frame2 = DataFrame(data, columns=['year', 'state',
'pop', 'debt'], index = ['one', 'two', 'three', 'four', 'five'])

>>> frame2
```

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# DATAFRAME CONTD

```
>>> frame2

>>> frame2.columns

>>> frame2.index

>>> frame2.values

>>> .head(10) #show the first n objects
```

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# PANDAS DATA ACCESS

- To access an element in a Series:
  - via integer index – s[i]
  - via label index – s[‘label’]
- To access a row in a dataframe:
  - df.loc[ [row name]](a dataframe), or df.loc[row name](a series)
  - df.iloc[[row no]](a dataframe), or df.iloc[row no] (a series)
- To access a column in a dataframe:
  - df[column name] (a series)
  - df[[column name]] (a dataframe)

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# PANDAS DATA ACCESS CONTD

- To access a cell in a dataframe
  - df.loc[row_name, column_name]
  - df.iloc[row_integer_index, column_integer_index]

- To access a segment in a dataframe
  - df.loc[[,],[,]] or df.iloc[[,],[,]]

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# MOVING DATA BETWEEN DATA STRUCTURES

- Quite often, we need to move data among between data structures. For example:
  - from a series to a list
    - ser.to_list()
  - from a dictionary to a dataframe
    - df.loc[].to_dict()

- See sample code for demonstration.



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